SCHOOL OF COMPUTING
Faculty of Engineering

Semester II 2019/2020

Subject : **Bioinformatics I (SCSB2103)**
Section : **01 – Dr Haslina Hashim**
Topic : **Lab 04 – BLAST**

KOO XUAN (A23CS0300)

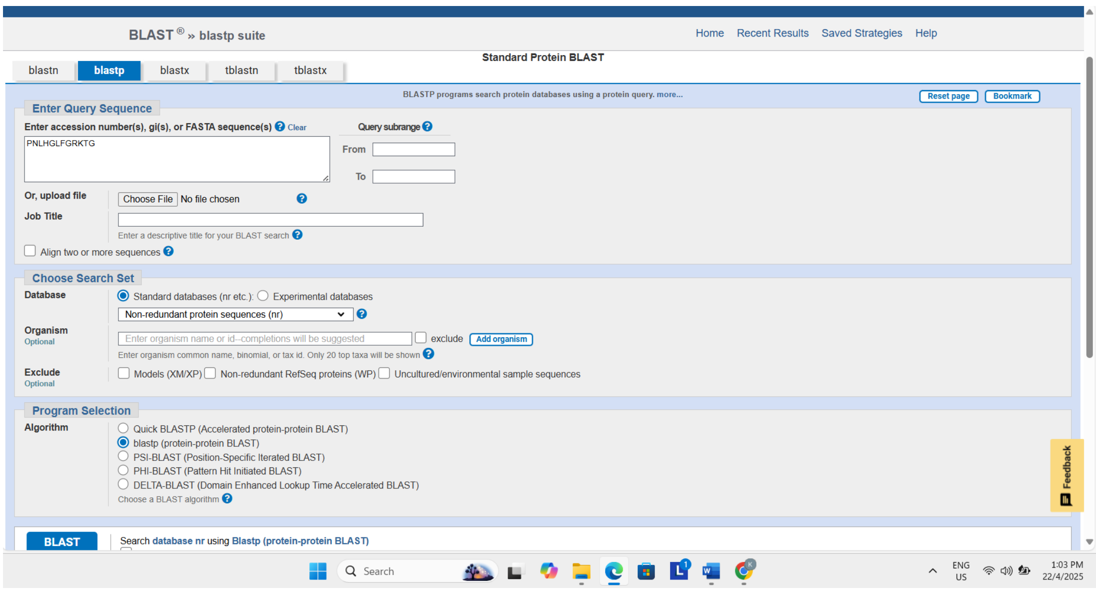
1) Perform a BLASTP search. In this problem we will explore the effect of a short protein query on the BLASTP parameters

   i) Perform a BLASTP search at NCBI using the following query of just 12 amino acids. By default, the parameters are adjusted for short queries.

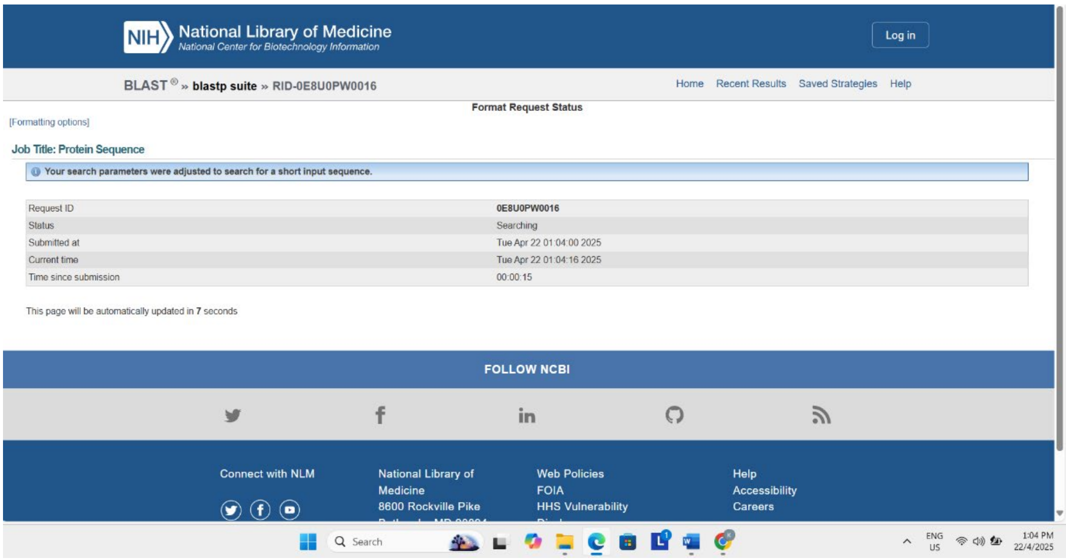
      (a) PNLHGLFGRKTG

   ii) Inspect the output. What is the E-value cut off? What is the word size? What is the scoring matrix? How do these settings compare to the default parameters?

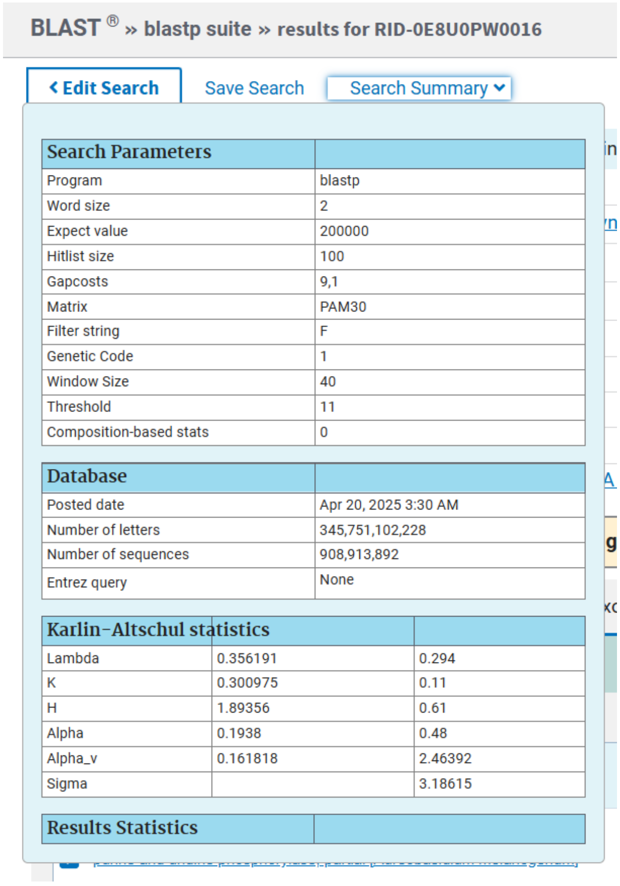
   First , we enter PNLHGLFGRKTG for the query sequence :



After click BLAST button, it will direct to here :



And automatically updated to here and we click search summary, and the results show:



BLAST ® » blastp suite » results for RID-0E8U0PW0016

Edit Search | Save Search | Search Summary

Search Parameters	
Program	blastp
Word size	2
Expect value	200000
Hitlist size	100
Gapcosts	9,1
Matrix	PAM30
Filter string	F
Genetic Code	1
Window Size	40
Threshold	11
Composition-based stats	0

Database	
Posted date	Apr 20, 2025 3:30 AM
Number of letters	345,751,102,228
Number of sequences	908,913,892
Entrez query	None

Karlin-Altschul statistics		
Lambda	0.356191	0.294
K	0.300975	0.11
H	1.89356	0.61
Alpha	0.1938	0.48
Alpha_v	0.161818	2.46392
Sigma		3.18615

Results Statistics	

The large Expect value serves the function of allowing me to see results in which the result may be biologically meaningful, but the Expect value (related to a probability value) was not very small. E could not be small because, given a short query, there is no opportunity for any alignment to achieve a high score (from E = kmne(exp)-lS a large score is associated with a small E value). The default Expect value is 10.

The matrix is PAM30 which is stringent relative to PAM250 or BLOSUM62 (the default matrix): mismatches are not well tolerated.

The word size is 2 although the default is 3.

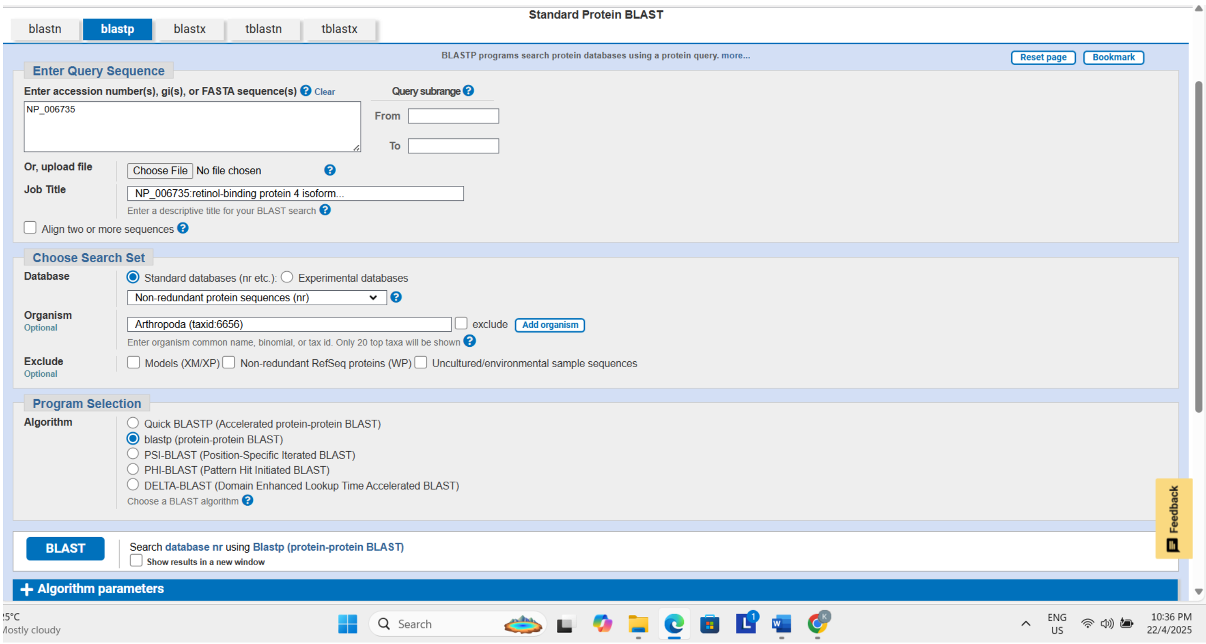
2) Protein searches are usually more informative than DNA searches. Why is this so? Do a BLASTP search

   i) Do a BLASTP search using RBP4 (NP_006735), restricting the output to Arthropoda (insects).

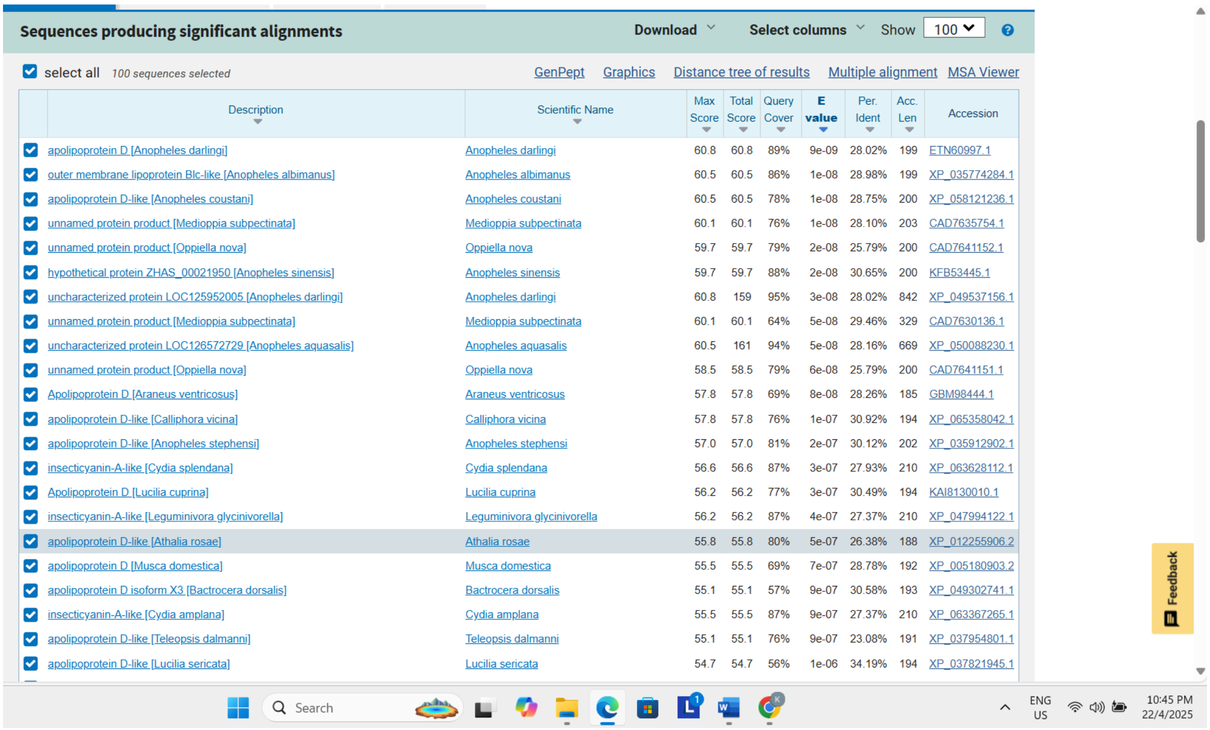
   ii) Next, do a BLASTN search using the RBP4 nucleotide sequence (NM_006744). For this query, select only the nucleotides corresponding to the coding region of the DNA. (To do this visit the NCBI Nucleotide page, follow the link to the coding sequence [CDS], then choose the FASTA format.)

   iii) Which search is more informative? How many databases matches have an E value less than 1.0 in each search?

Set the settings for the BLASTP search:

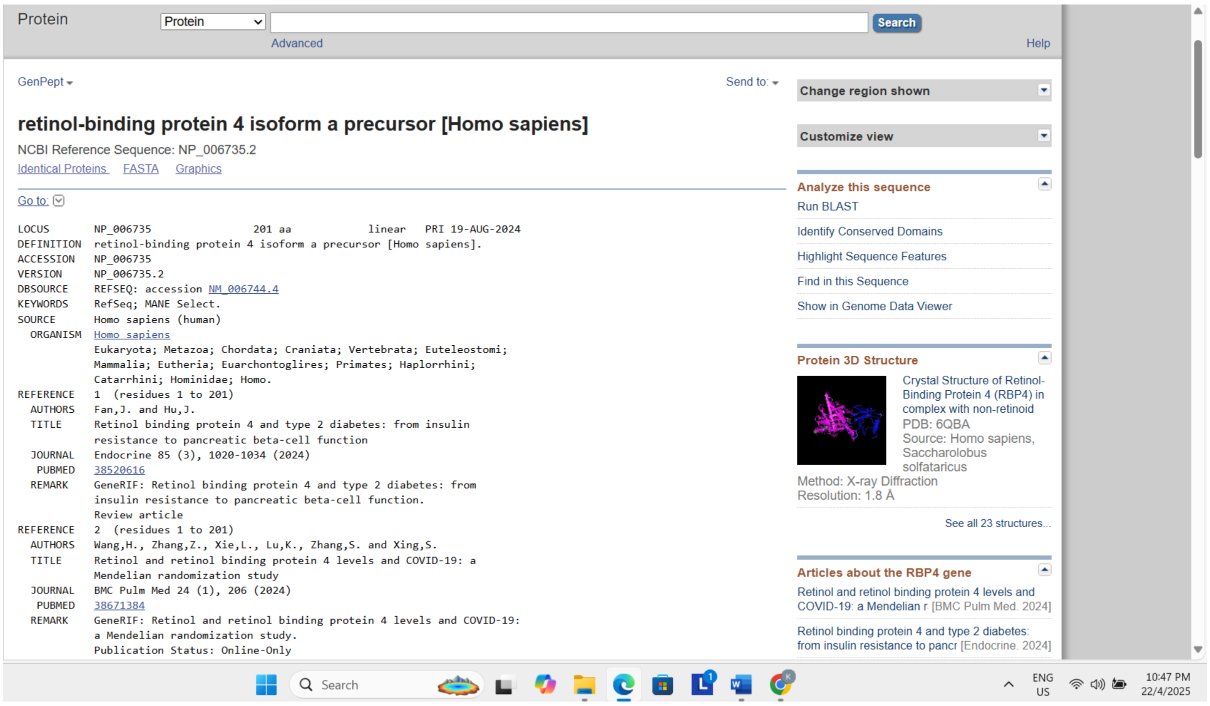


There are about 100 matches having an E value less than 0.01 (while the problem asks about an E value <1, it would have been better to ask about E<0.01). Here are the first matches:

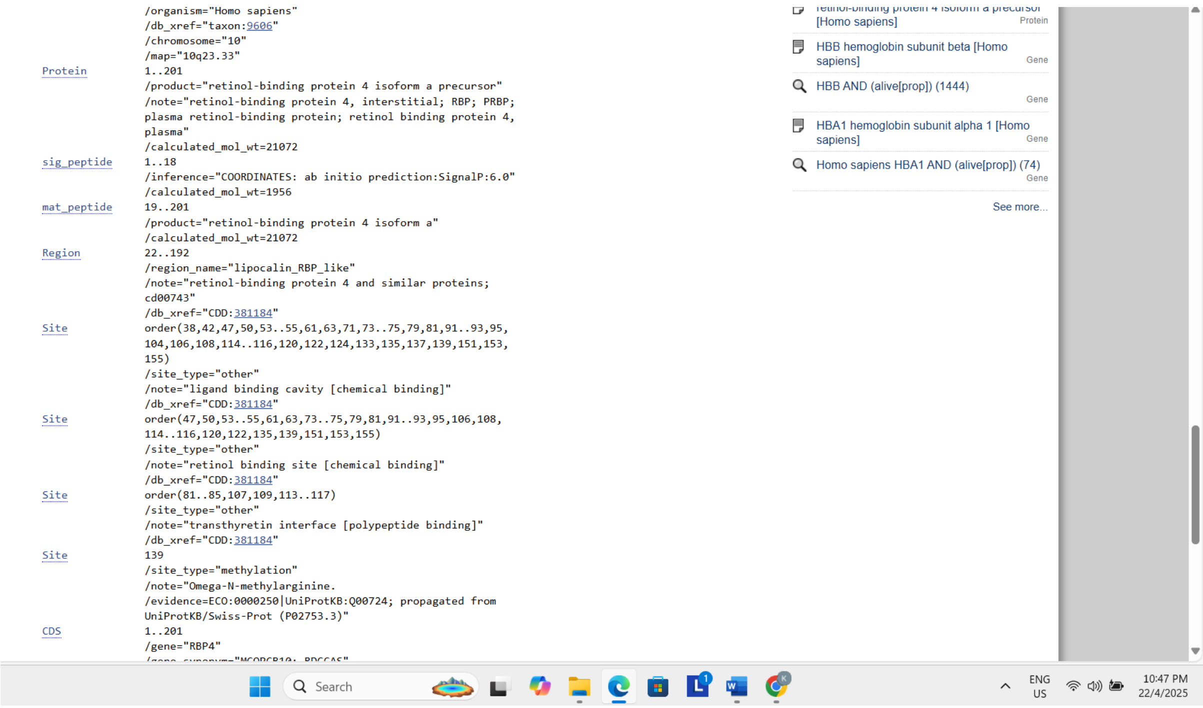


Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
apolipoprotein D [Anopheles darlingi]	Anopheles darlingi	60.8	60.8	89%	9e-09	28.02%	199	ETN60997.1
outer membrane lipoprotein Blc-like [Anopheles albimanus]	Anopheles albimanus	60.5	60.5	86%	1e-08	28.98%	199	XP_035774284.1
apolipoprotein D-like [Anopheles coustani]	Anopheles coustani	60.5	60.5	78%	1e-08	28.75%	200	XP_058121236.1
unnamed protein product [Medioppia subpectinata]	Medioppia subpectinata	60.1	60.1	76%	1e-08	28.10%	203	CAD7635754.1
unnamed protein product [Oppiella nova]	Oppiella nova	59.7	59.7	79%	2e-08	25.79%	200	CAD7641152.1
hypothetical protein ZHAS_00021950 [Anopheles sinensis]	Anopheles sinensis	59.7	59.7	88%	2e-08	30.65%	200	KFB53445.1
uncharacterized protein LOC125952005 [Anopheles darlingi]	Anopheles darlingi	60.8	159	95%	3e-08	28.02%	842	XP_049537156.1
unnamed protein product [Medioppia subpectinata]	Medioppia subpectinata	60.1	60.1	64%	5e-08	29.46%	329	CAD7630136.1
uncharacterized protein LOC126572729 [Anopheles aquasalis]	Anopheles aquasalis	60.5	161	94%	5e-08	28.16%	669	XP_050088230.1
unnamed protein product [Oppiella nova]	Oppiella nova	58.5	58.5	79%	6e-08	25.79%	200	CAD7641151.1
Apolipoprotein D [Araneus ventricosus]	Araneus ventricosus	57.8	57.8	69%	8e-08	28.26%	185	GBM98444.1
apolipoprotein D-like [Calliphora vicina]	Calliphora vicina	57.8	57.8	76%	1e-07	30.92%	194	XP_065358042.1
apolipoprotein D-like [Anopheles stephensi]	Anopheles stephensi	57.0	57.0	81%	2e-07	30.12%	202	XP_035912902.1
insecticyanin-A-like [Cydia splendana]	Cydia splendana	56.6	56.6	87%	3e-07	27.93%	210	XP_063628112.1
Apolipoprotein D [Lucilia cuprina]	Lucilia cuprina	56.2	56.2	77%	3e-07	30.49%	194	KAI8130010.1
insecticyanin-A-like [Leguminivora glycinivorella]	Leguminivora glycinivorella	56.2	56.2	87%	4e-07	27.37%	210	XP_047994122.1
apolipoprotein D-like [Athalia rosae]	Athalia rosae	55.8	55.8	80%	5e-07	26.38%	188	XP_012255906.2
apolipoprotein D [Musca domestica]	Musca domestica	55.5	55.5	69%	7e-07	28.78%	192	XP_005180903.2
apolipoprotein D isoform X3 [Bactrocera dorsalis]	Bactrocera dorsalis	55.1	55.1	57%	9e-07	30.58%	193	XP_049302741.1
insecticyanin-A-like [Cydia amplana]	Cydia amplana	55.5	55.5	87%	9e-07	27.37%	210	XP_063367265.1
apolipoprotein D-like [Teleopsis dalmanni]	Teleopsis dalmanni	55.1	55.1	76%	9e-07	23.08%	191	XP_037954801.1
apolipoprotein D-like [Lucilia sericata]	Lucilia sericata	54.7	54.7	56%	1e-06	34.19%	194	XP_037821945.1

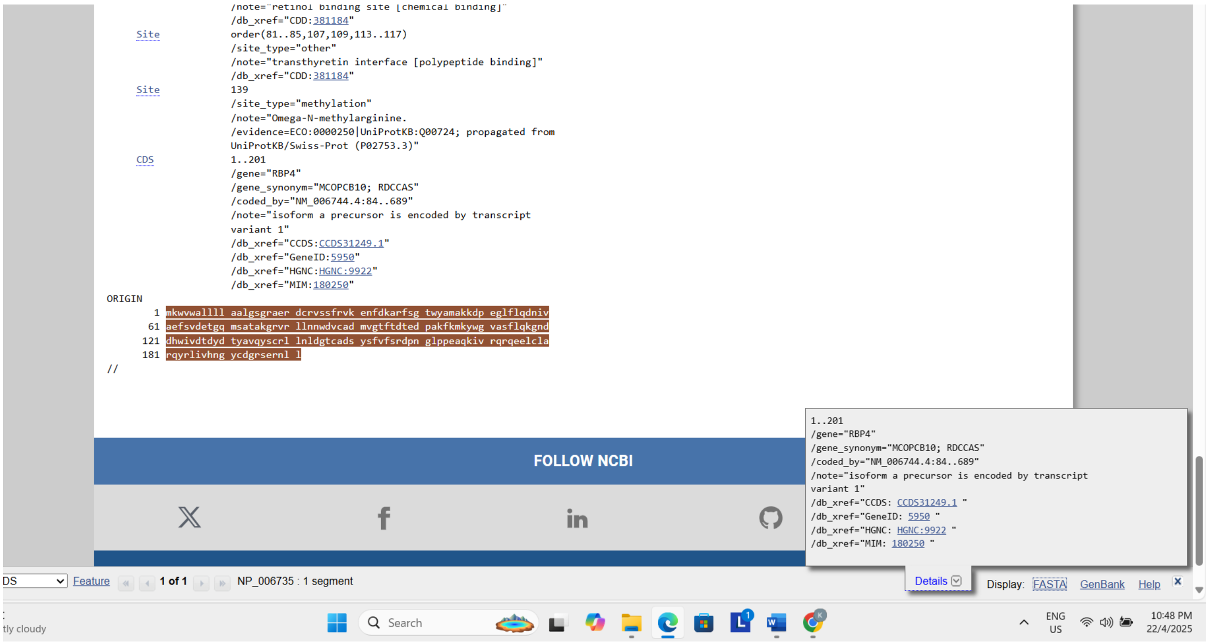
Next find the DNA coding sequence. The BLASTP result has a link to the NCBI protein page; this page (http://www.ncbi.nlm.nih.gov/protein/NP_006735.2) has a link to the DNA (NM_006744.3):

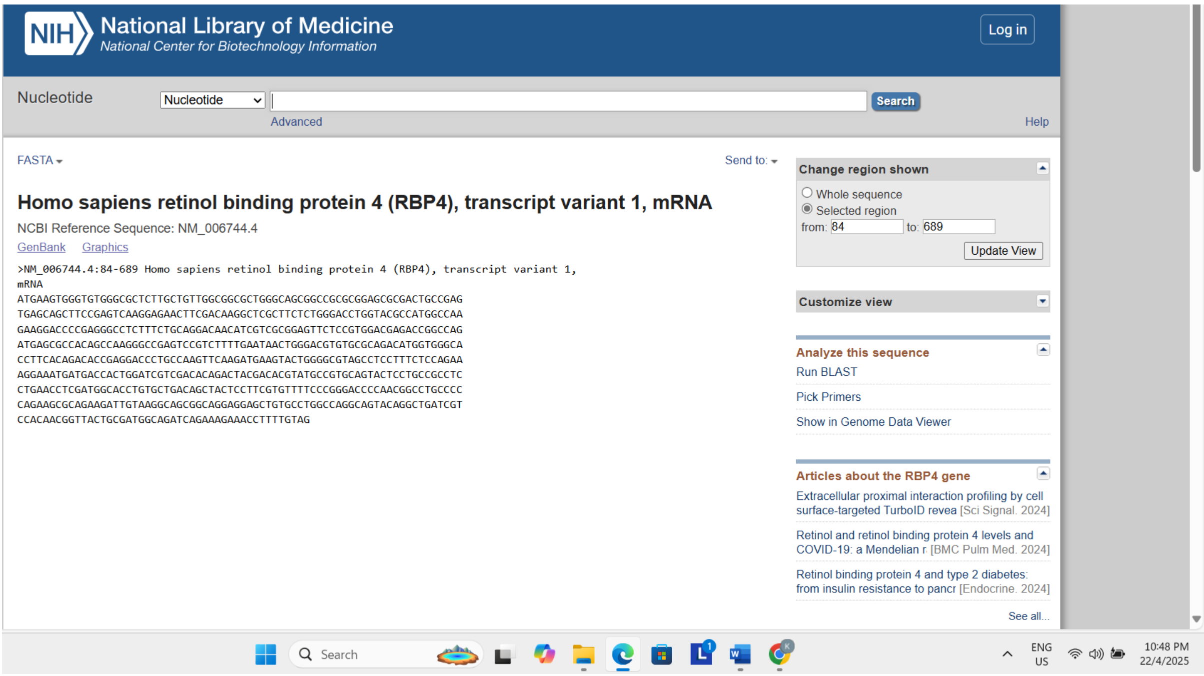


This is the features, from the features find CDS:

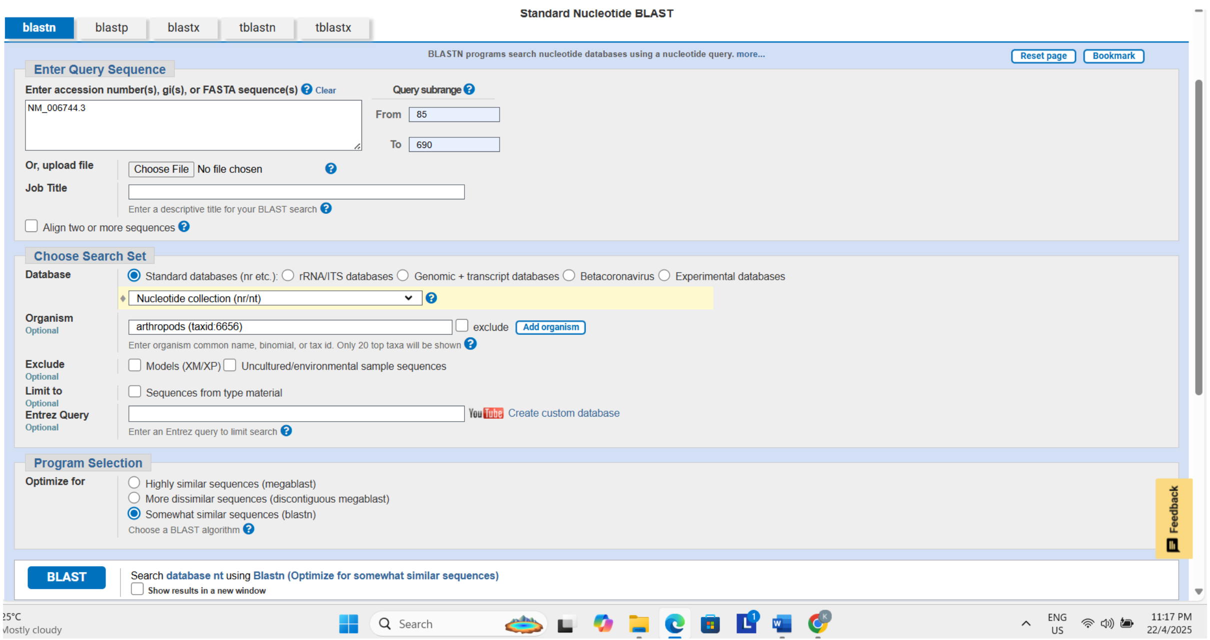


CDS is highlighted. At the bottom of the page there is a link to FASTA which produces the sequence starting ATG and ending with a stop codon:

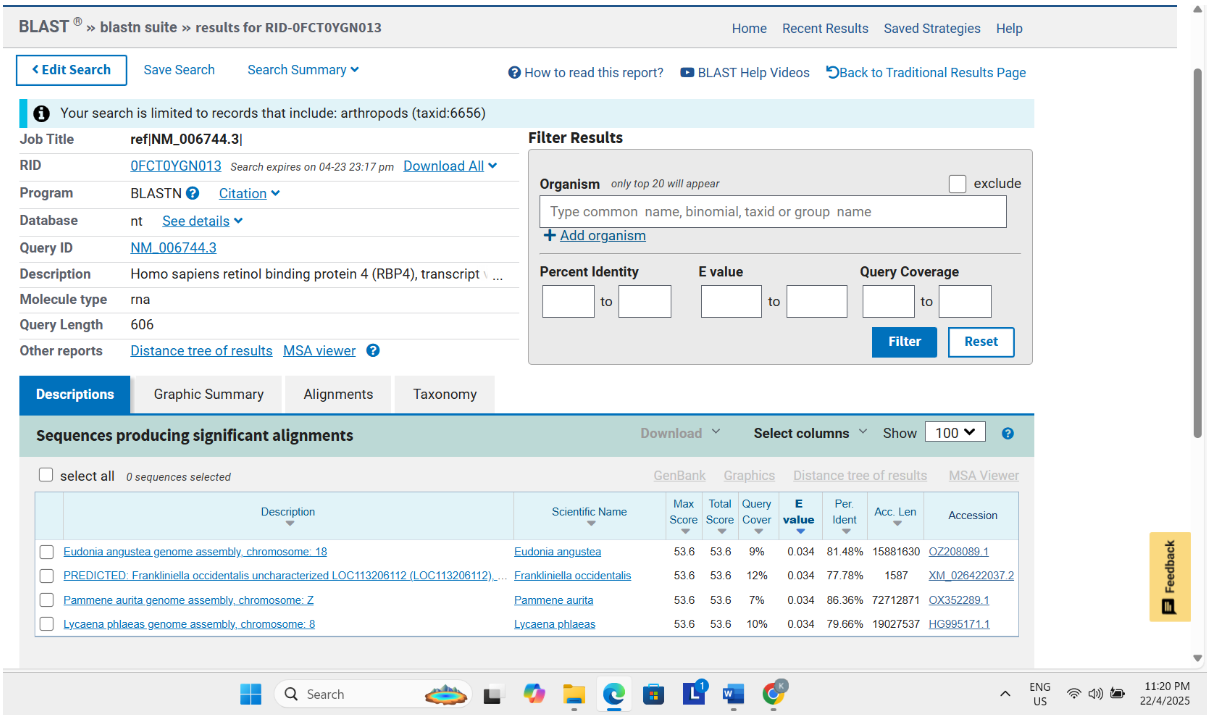




Use this as a query in a BLASTN search. Note the sequence range is automatically specified to correspond to the CDS:



The result is as follows:



There are no database matches having an Expect value <0.01. The protein search is far more sensitive than the DNA search.

3) This problem introduces batch queries. It is possible to search many queries simultaneously, either using the web-based BLAST (as in this problem) or via locally installed BLAST+.

   Mosses are plants of the phylum Bryophyta, including the non-seed plant Physcomitrella patens that had its genome sequenced (Rensing et al., 2008).

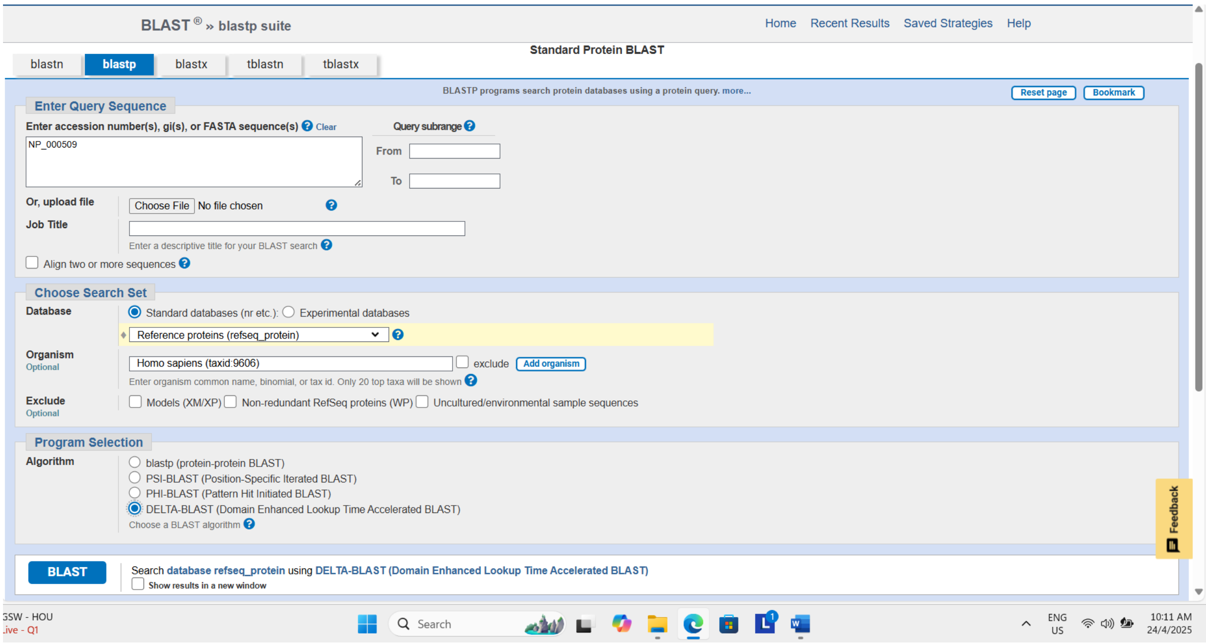
   Do mosses have any globin proteins, and if so, which human globin(s) are they most closely related to?

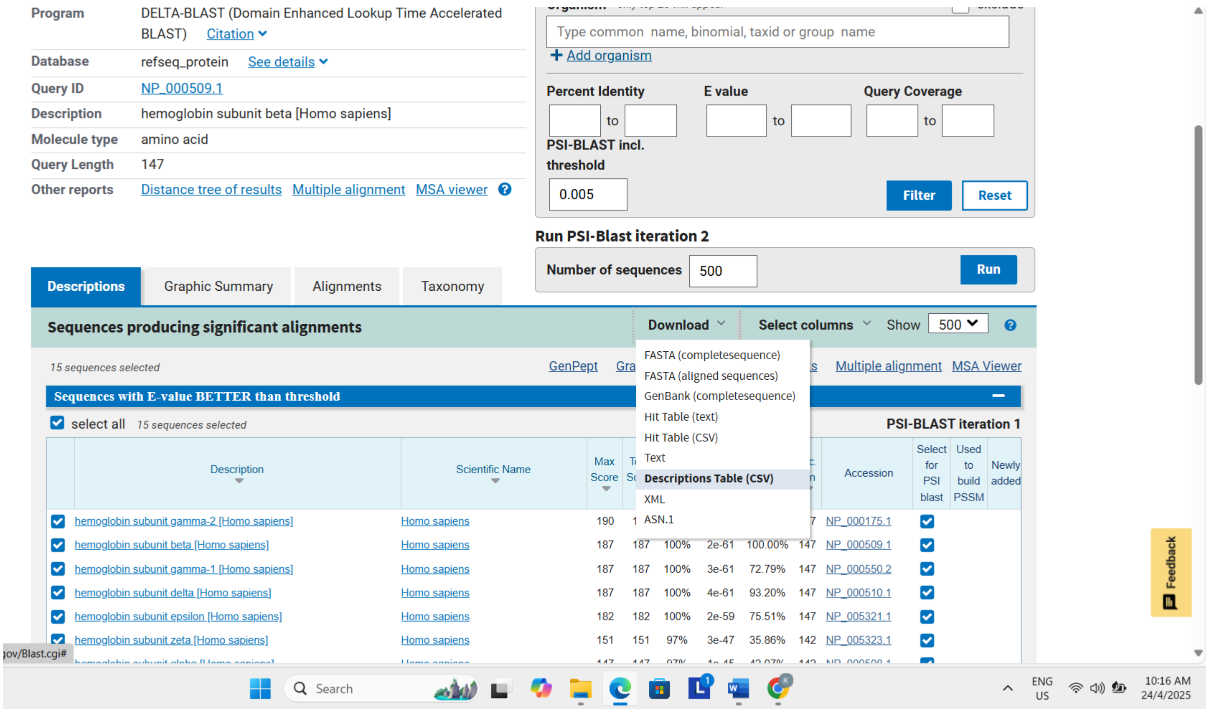
i) First obtain the accession numbers of all human globins. There are several approaches to doing this, including BLASTP using beta globin and neuroglobin as queries. Other approaches involve DELTA-BLAST (Chapter 5), or Pfam (Chapter 6). These accession numbers are provided in Web Document 4.7.

ii) Perform a BLASTP search using all accession numbers as queries, entering them into the query box. Restrict the output to RefSeq proteins of the mosses.

iii) Results for each query are shown (one at a time) via a pull-down menu. Currently there are significant, although distant matches of all human globins to moss proteins except for hemoglobin subunit mu. (See for example the match between human epsilon globin and predicted moss protein XP_001786089.1 with an E value of 0.01. A BLASTP search with that moss protein confirms it is related to many annotated plant globins.) Notably, only one human protein (neuroglobin, NP_001030585.1) has very strong matches to moss proteins such as P. patens predicted protein XP_001764902.1 (E value 2e-10, 27% identity across a span of 138 amino acid residues).

I have obtained a list of accession numbers of human RefSeq proteins that are globins by using HBB (NP_000509) in a DELTA-BLAST search, selecting "all" of the 15 significant results, clicking "download" and sending the results to a csv file, and parsing them to get this list:

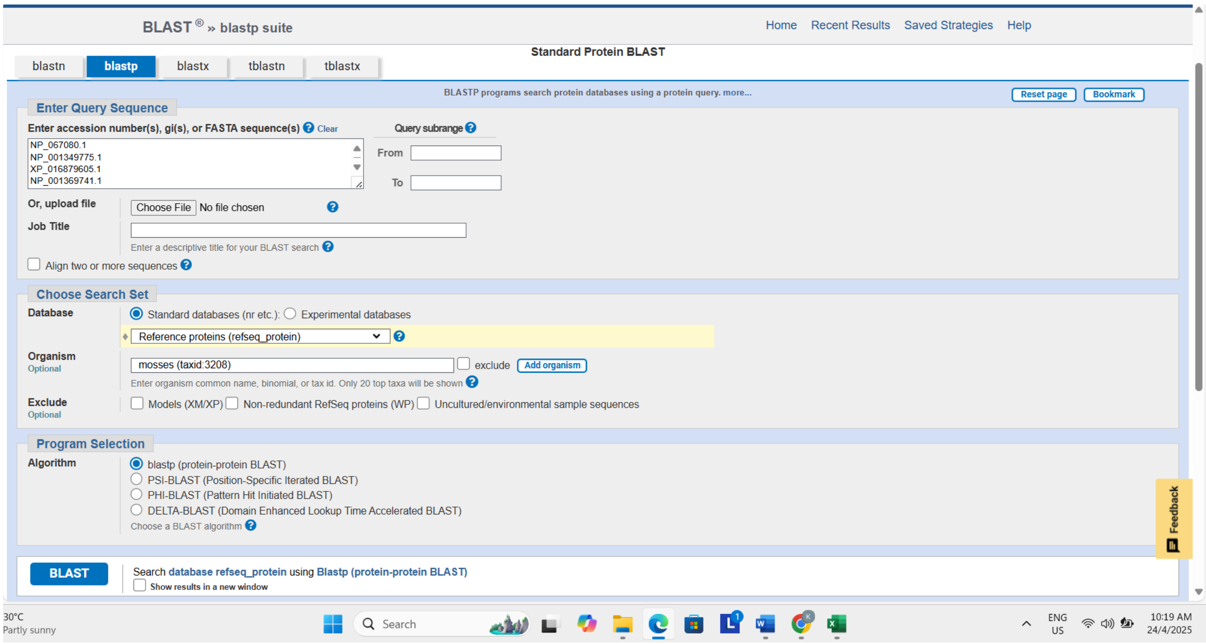




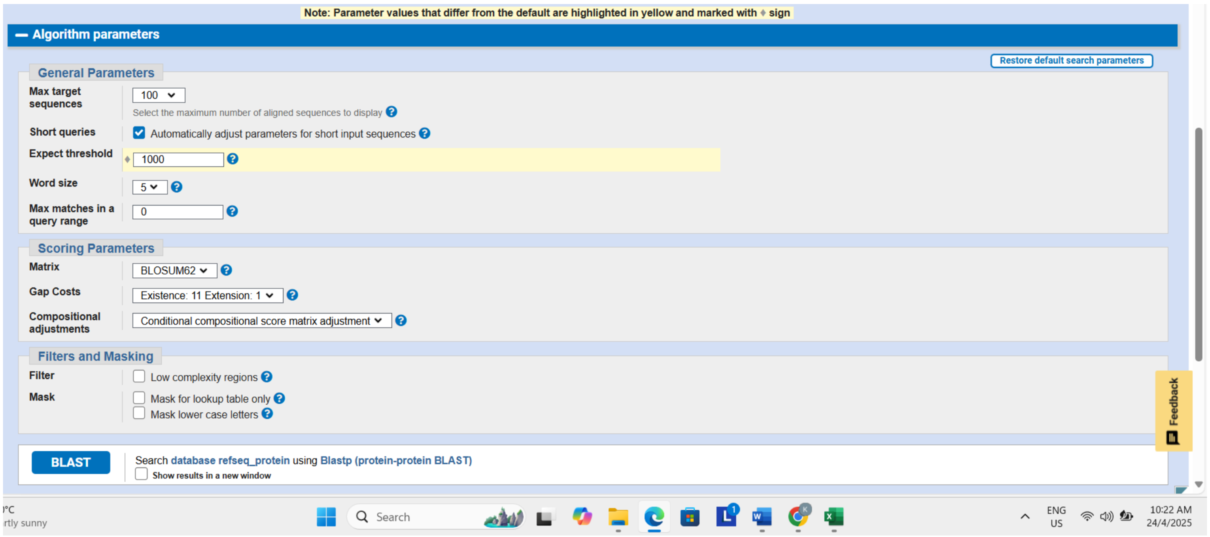
And get the accession:

NP_000175.1
NP_000509.1
NP_000550.2
NP_000510.1
NP_005321.1
NP_005323.1
NP_000508.1
NP_005322.1
NP_001003938.1
XP_005257062.1
NP_599030.1
NP_067080.1
NP_001349775.1
XP_016879605.1
NP_001369741.1

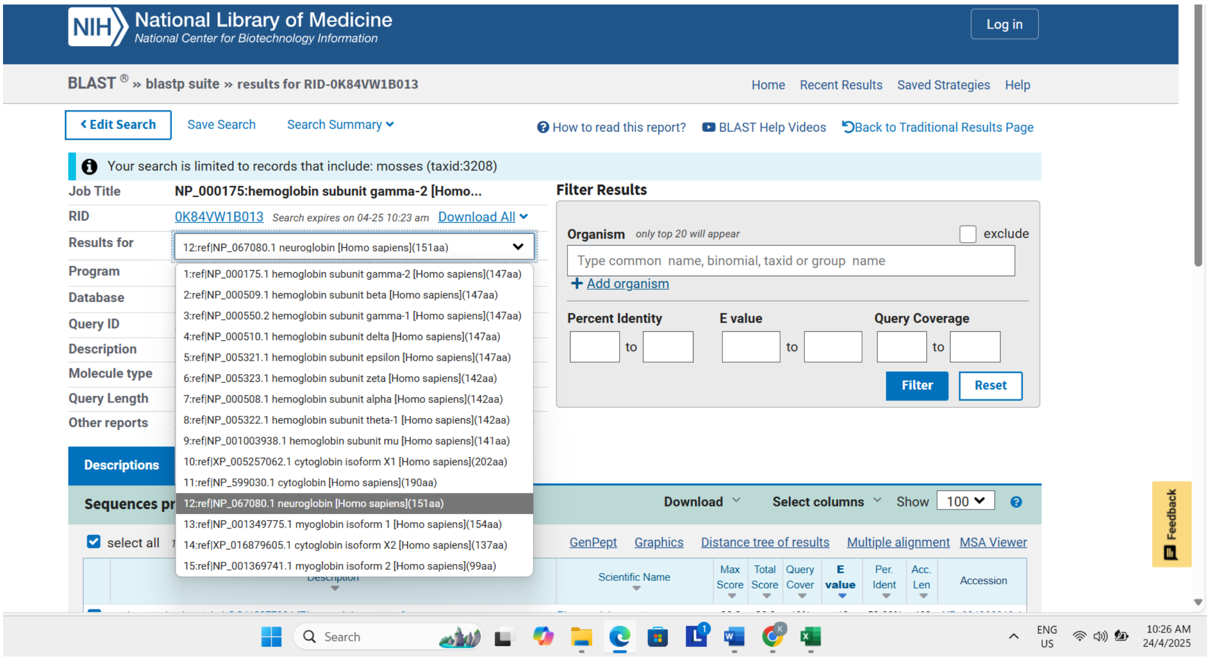
Then performed a BLASTP search of the RefSeq database, restricting the organism to mosses.



And edit the algorithm parameters:



Here the output and it includes a pull-down menu allowing you to inspect the results from each protein query. The best results (shown below) are for human neuroglobin as a query; matches to moss proteins have convincing Expect values of 7.6,14 and 18.


National Library of Medicine
National Center for Biotechnology Information
Log in
BLAST ® » blastp suite » results for RID-0K84VW1B013
Home Recent Results Saved Strategies Help
Edit Search Save Search Search Summary
How to read this report? BLAST Help Videos Back to Traditional Results Page
Your search is limited to records that include: mosses (taxid:3208)
Job Title NP_000175:hemoglobin subunit gamma-2 [Homo...
RID 0K84VW1B013 Search expires on 04-25 10:23 am Download All
Results for 12:ref|NP_067080.1 neuroglobin [Homo sapiens](151aa)
Program
Database
Query ID
Description
Molecule type
Query Length
Other reports
1:ref|NP_000175.1 hemoglobin subunit gamma-2 [Homo sapiens](147aa)
2:ref|NP_000509.1 hemoglobin subunit beta [Homo sapiens](147aa)
3:ref|NP_000550.2 hemoglobin subunit gamma-1 [Homo sapiens](147aa)
4:ref|NP_000510.1 hemoglobin subunit delta [Homo sapiens](147aa)
5:ref|NP_005321.1 hemoglobin subunit epsilon [Homo sapiens](147aa)
6:ref|NP_005323.1 hemoglobin subunit zeta [Homo sapiens](142aa)
7:ref|NP_000508.1 hemoglobin subunit alpha [Homo sapiens](142aa)
8:ref|NP_005322.1 hemoglobin subunit theta-1 [Homo sapiens](142aa)
9:ref|NP_001003938.1 hemoglobin subunit mu [Homo sapiens](141aa)
10:ref|XP_005257062.1 cytoglobin isoform X1 [Homo sapiens](202aa)
11:ref|NP_599030.1 cytoglobin [Homo sapiens](190aa)
12:ref|NP_067080.1 neuroglobin [Homo sapiens](151aa)
13:ref|NP_001349775.1 myoglobin isoform 1 [Homo sapiens](154aa)
14:ref|XP_016879605.1 cytoglobin isoform X2 [Homo sapiens](137aa)
15:ref|NP_001369741.1 myoglobin isoform 2 [Homo sapiens](99aa)
Filter Results
Organism only top 20 will appear exclude
Type common name, binomial, taxid or group name
Add organism
Percent Identity E value Query Coverage
to to to
Filter Reset
Descriptions
Sequences pr
Download Select columns Show 100
select all
GenPept Graphics Distance tree of results Multiple alignment MSA Viewer
Scientific Name Max Score Total Score Query Cover E value Per. Ident Acc. Len Accession
Feedback
Search
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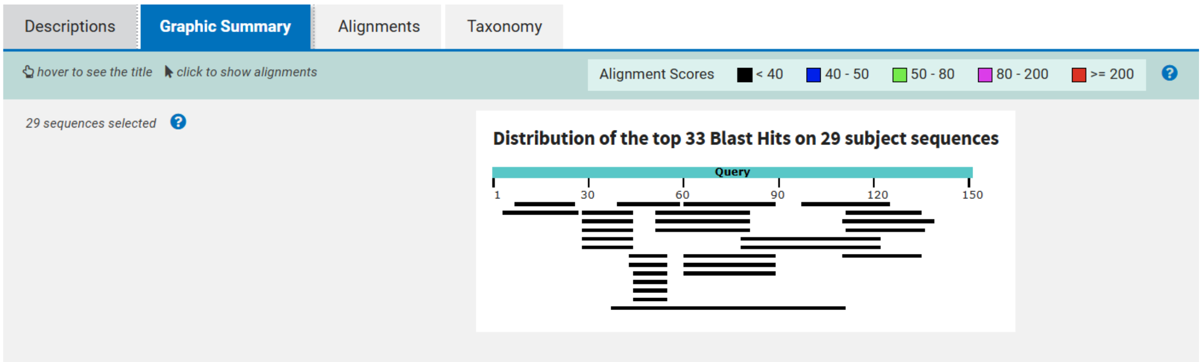
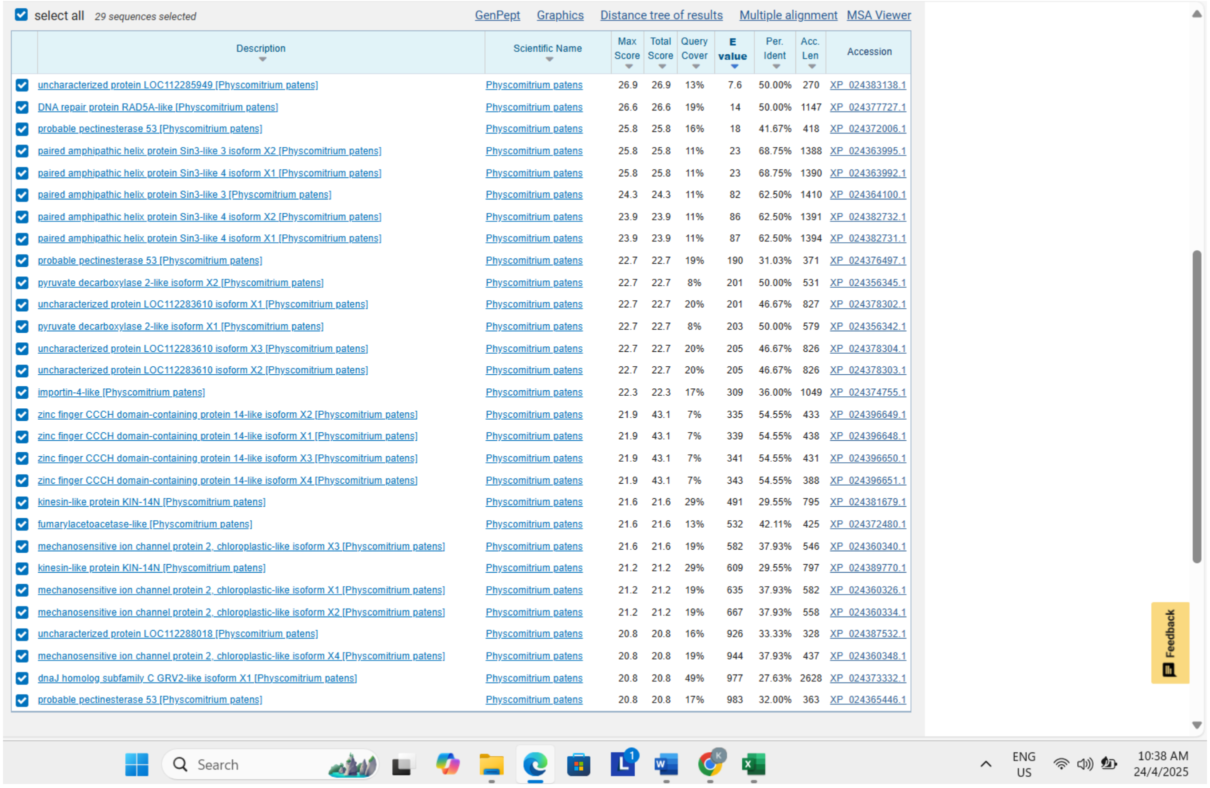

Descriptions Graphic Summary Alignments Taxonomy
hover to see the title click to show alignments
Alignment Scores < 40 40 - 50 50 - 80 80 - 200 >= 200
29 sequences selected
Distribution of the top 33 Blast Hits on 29 subject sequences
Query
1 30 60 90 120 150



select all 29 sequences selected GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☑	uncharacterized protein LOC112285949 [Physcomitrium patens]	Physcomitrium patens	26.9	26.9	13%	7.6	50.00%	270	XP_024383138.1
☑	DNA repair protein RAD5A-like [Physcomitrium patens]	Physcomitrium patens	26.6	26.6	19%	14	50.00%	1147	XP_024377727.1
☑	probable pectinesterase 53 [Physcomitrium patens]	Physcomitrium patens	25.8	25.8	16%	18	41.67%	418	XP_024372006.1
☑	paired amphipathic helix protein Sin3-like 3 isoform X2 [Physcomitrium patens]	Physcomitrium patens	25.8	25.8	11%	23	68.75%	1388	XP_024363995.1
☑	paired amphipathic helix protein Sin3-like 4 isoform X1 [Physcomitrium patens]	Physcomitrium patens	25.8	25.8	11%	23	68.75%	1390	XP_024363992.1
☑	paired amphipathic helix protein Sin3-like 3 [Physcomitrium patens]	Physcomitrium patens	24.3	24.3	11%	82	62.50%	1410	XP_024364100.1
☑	paired amphipathic helix protein Sin3-like 4 isoform X2 [Physcomitrium patens]	Physcomitrium patens	23.9	23.9	11%	86	62.50%	1391	XP_024382732.1
☑	paired amphipathic helix protein Sin3-like 4 isoform X1 [Physcomitrium patens]	Physcomitrium patens	23.9	23.9	11%	87	62.50%	1394	XP_024382731.1
☑	probable pectinesterase 53 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	19%	190	31.03%	371	XP_024376497.1
☑	pyruvate decarboxylase 2-like isoform X2 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	8%	201	50.00%	531	XP_024356345.1
☑	uncharacterized protein LOC112283610 isoform X1 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	20%	201	46.67%	827	XP_024378302.1
☑	pyruvate decarboxylase 2-like isoform X1 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	8%	203	50.00%	579	XP_024356342.1
☑	uncharacterized protein LOC112283610 isoform X3 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	20%	205	46.67%	826	XP_024378304.1
☑	uncharacterized protein LOC112283610 isoform X2 [Physcomitrium patens]	Physcomitrium patens	22.7	22.7	20%	205	46.67%	826	XP_024378303.1
☑	importin-4-like [Physcomitrium patens]	Physcomitrium patens	22.3	22.3	17%	309	36.00%	1049	XP_024374755.1
☑	zinc finger CCCH domain-containing protein 14-like isoform X2 [Physcomitrium patens]	Physcomitrium patens	21.9	43.1	7%	335	54.55%	433	XP_024396649.1
☑	zinc finger CCCH domain-containing protein 14-like isoform X1 [Physcomitrium patens]	Physcomitrium patens	21.9	43.1	7%	339	54.55%	438	XP_024396648.1
☑	zinc finger CCCH domain-containing protein 14-like isoform X3 [Physcomitrium patens]	Physcomitrium patens	21.9	43.1	7%	341	54.55%	431	XP_024396650.1
☑	zinc finger CCCH domain-containing protein 14-like isoform X4 [Physcomitrium patens]	Physcomitrium patens	21.9	43.1	7%	343	54.55%	388	XP_024396651.1
☑	kinesin-like protein KIN-14N [Physcomitrium patens]	Physcomitrium patens	21.6	21.6	29%	491	29.55%	795	XP_024381679.1
☑	fumarylacetoacetase-like [Physcomitrium patens]	Physcomitrium patens	21.6	21.6	13%	532	42.11%	425	XP_024372480.1
☑	mechanosensitive ion channel protein 2, chloroplastic-like isoform X3 [Physcomitrium patens]	Physcomitrium patens	21.6	21.6	19%	582	37.93%	546	XP_024360340.1
☑	kinesin-like protein KIN-14N [Physcomitrium patens]	Physcomitrium patens	21.2	21.2	29%	609	29.55%	797	XP_024389770.1
☑	mechanosensitive ion channel protein 2, chloroplastic-like isoform X1 [Physcomitrium patens]	Physcomitrium patens	21.2	21.2	19%	635	37.93%	582	XP_024360326.1
☑	mechanosensitive ion channel protein 2, chloroplastic-like isoform X2 [Physcomitrium patens]	Physcomitrium patens	21.2	21.2	19%	667	37.93%	558	XP_024360334.1
☑	uncharacterized protein LOC112288018 [Physcomitrium patens]	Physcomitrium patens	20.8	20.8	16%	926	33.33%	328	XP_024387532.1
☑	mechanosensitive ion channel protein 2, chloroplastic-like isoform X4 [Physcomitrium patens]	Physcomitrium patens	20.8	20.8	19%	944	37.93%	437	XP_024360348.1
☑	dnaJ homolog subfamily C GRV2-like isoform X1 [Physcomitrium patens]	Physcomitrium patens	20.8	20.8	49%	977	27.63%	2628	XP_024373332.1
☑	probable pectinesterase 53 [Physcomitrium patens]	Physcomitrium patens	20.8	20.8	17%	983	32.00%	363	XP_024365446.1

4) Is the pol protein of HIV-1 more closely related to the pol protein of HIV-2 or to the pol protein of simian immunodeficiency virus (SIV)? Use the BLASTP program to decide.

Hint: try the Entrez command “NOT hiv-1[organism]” to focus the search away from HIV-1 matches.

Go to website (http://www.ncbi.nlm.nih.gov/protein/28872819?report=fasta) and click on protein click on fasta to get this:

```
>NP_057849.4 Gag-Pol [Human immunodeficiency virus 1]
MGARASVLSGGELDRWEKIRLRPGGKKKYKLKHIVWASRELERFAVNPGLLETSEGCRQILGQLQPSLQT
GSEELRSLYNTVATLYCVHQRIEIKDTKEALDKIEEEQNKSKKKAQQAAADTGHSNQVSQNYPIVQNIQG
QMVHQAISPRTLNAWVKVVEEKAFSPEVIPMFSALSEGATPQDLNTMLNTVGGHQAAMQMLKETINEEAA
EWDRVHPVHAGPIAPGQMREPRGSDIAGTTSTLQEQIGWMTNNPPIPVGEIYKRWIILGLNKIVRMYSPT
SILDIRQGPKEPFRDYVDRFYKTLRAEQASQEVKNWMTETLLVQNANPDCKTILKALGPAATLEEMMTAC
QGVGGPGHKARVLAEAMSQVTNSATIMMQRGNFRNQRKIVKCFNCGKEGHTARNCRAPRKKGCWKCGKEG
HQMKDCTERQANFLREDLAFLQGKAREFSSEQTRANSPTRRELQVWGRDNNSPSEAGADRQGTVSFNFPQ
VTLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAI
GTVLVGPTPVNIIGRNLLTQIGCTLNFPISPIETVPVKLKPGMDGPKVKQWPLTEEKIKALVEICTEMEK
EGKISKIGPENPYNTPVFAIKKKDSTKWRKLVDFRELNKRTQDFWEVQLGIPHPAGLKKKKSVTVLDVGD
AYFSVPLDEDFRKYTAFTIPSINNETPGIRYQYNVLPQGWKGSPAIFQSSMTKILEPFRKQNPDIVIYQY
MDDLYVGSDLEIGQHRTKIEELRQHLLRWGLTTPDKKHQKEPPFLWMGYELHPDKWTVQPIVLPEKDSWT
VNDIQKLVGKLNWASQIYPGIKVRQLCKLLRGTKALTEVIPLTEEAELELAENREILKEPVHGVYYDPSK
DLIAEIQKQGQGQWTYQIYQEPFKNLKTGKYARMRGAHTNDVKQLTEAVQKITTESIVIWGKTPKFKLPI
QKETWETWWTEYWQATWIPEWEFVNTPPLVKLWYQLEKEPIVGAETFYVDGAANRETKLGKAGYVTNRGR
```

QKVVTLTDTTNQKTELQAIYLALQDSGLEVNIVTDSQYALGIIQAQPDQSESELVNQIIEQLIKKEKVYL
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QLKGEAMHGQVDCSPGIWQLDCTHLEGKVILVAVHVASGYIEAEVIPAETGQETAYFLLKLAGRWPVKTI
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NFKRKGGIGGYSAGERIVDIIATDIQTKELQKQITKIQNFRVYYRDSRNPLWKGPAKLLWKGEGAVVIQD
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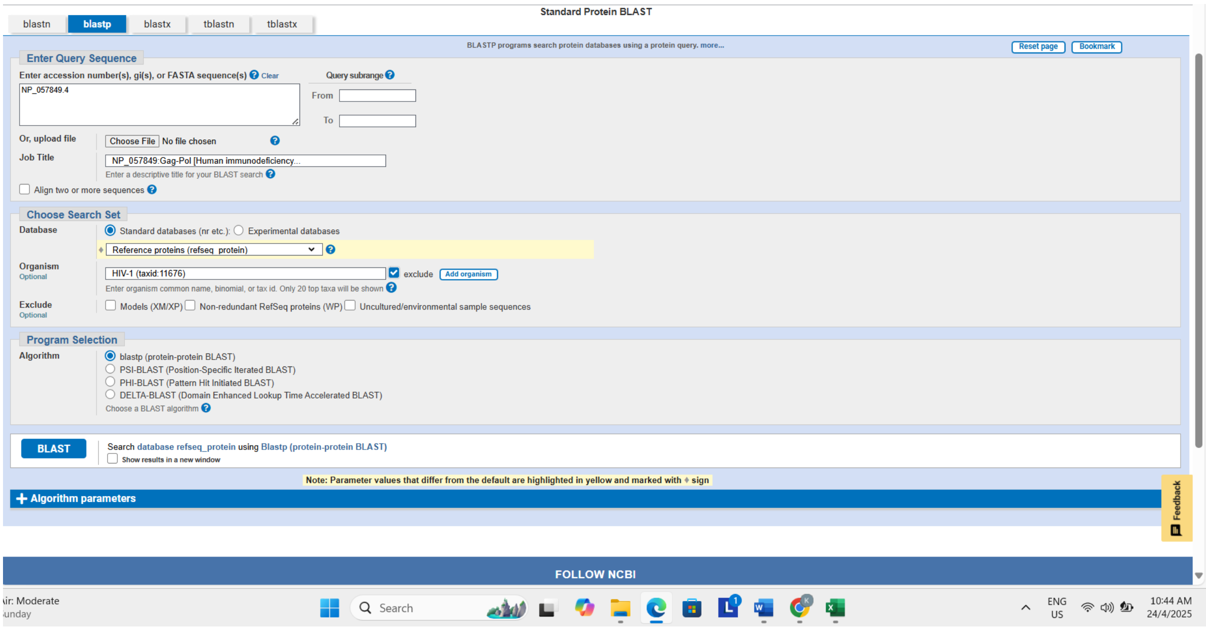
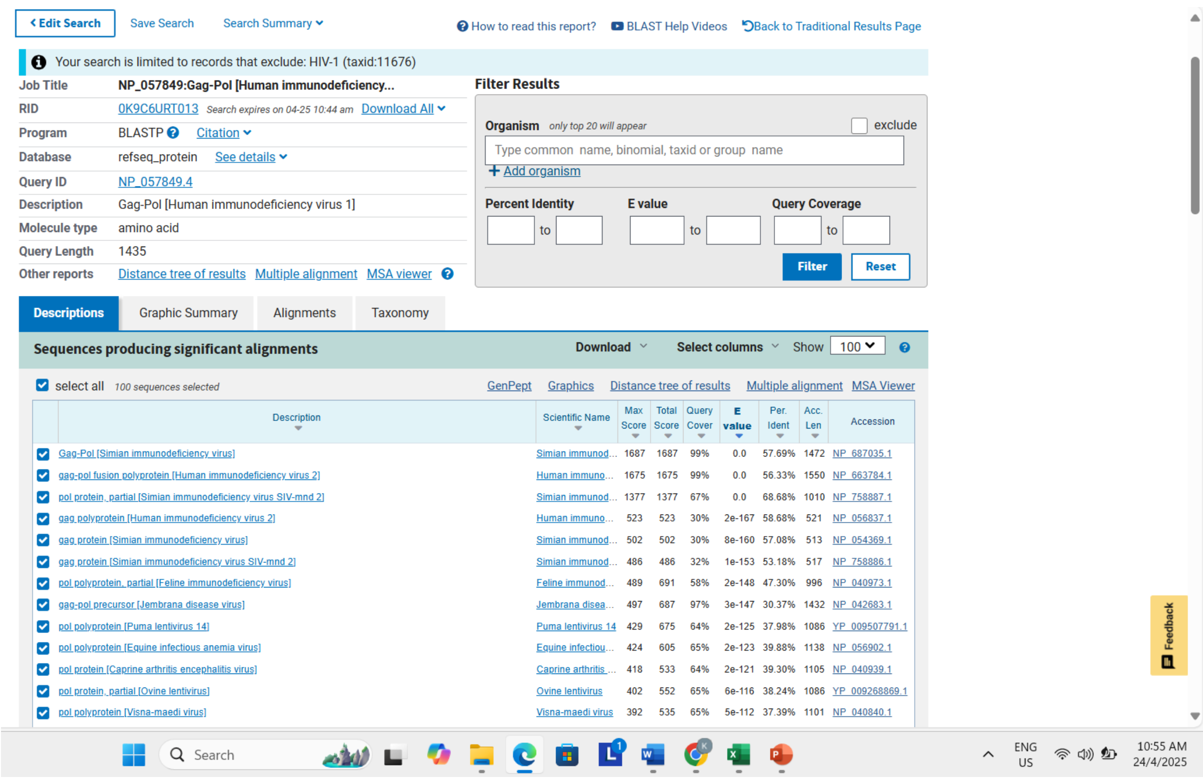


Here is the result:



Both proteins are comparably related. The SIV protein and HIV 2 has the same higher coverage (99% versus 99%) and both have E values of 0.

5) You perform a BLAST search and a result has an E value of about 1 × 10–4. What does this E value mean? What are some parameters on which an E value depends?

An E value (Expect value) of 1×10^{-4} in a BLAST search means that the probability of obtaining an alignment with a similar or better score purely by chance is 1 in 10,000. This suggests that the match is statistically significant and unlikely to have occurred randomly. Therefore, the null hypothesis that the alignment occurred by chance can be rejected, supporting the idea that the sequences are homologous. The E value depends on several parameters, including the alignment score, the length of the query sequence, the size of the database searched, the substitution matrix used, and the gap penalties, all of which influence how alignment scores are calculated.